

Fraud Detection with Deep Learning: Interpretable Transaction Monitoring

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OBJECTIVES

- Primary Goal:** Build and evaluate a robust deep learning architecture using PyTorch to detect fraudulent activities within highly imbalanced transaction streams, maximizing the PR-AUC metric.
- Can a fully connected Multi-Layer Perceptron (MLP) outperform traditional tree-based architectures under extreme class imbalance?
 - Which contextual transaction patterns contribute most heavily to automated risk escalation?
 - Core Innovation:** Blending local and global SHAP explanations with a custom optimized neural network to deliver reliable, auditable risk insight.

METHODOLOGY

Dataset & Imbalance: Evaluated on a financial transaction corpus exceeding 284,000 records. Class imbalance is severe, with a fraud ratio of exactly **1:200** (0.17% fraud rate).

Preprocessing Pipeline: Standard scaling is computed strictly within training boundaries to completely isolate validation sets. Imputation and stratified **70/15/15** data splitting routines are structurally defined within `data_loader.py`.

Model Architecture: Feedforward multi-layer perceptron topology: **Dense 128 → 64 → 32 → Sigmoid**. Integrated with robust ReLU activations and active Dropout layers fixed at 0.3 to systematically neutralize overfitting threats.

Training Details: Optimized via Adam ($\text{lr}=1\text{e-}3$), a batch size of 256, and Early Stopping capped at 50 epochs (patience=7). Class skew is addressed via a weighted BCELoss layer utilizing a calibrated `pos_weight` multiplier.

```
=== FRAUD DETECTION - Manual Transaction Entry ===
Amount ($): 100
Available ($): 5000
Balance ($): 1200
Credit Limit ($): 6000
Card Present: No
CVV Match: Yes
Analyse Transaction

=== TRANSACTION RISK REPORT ===
Reconstruction Error : 0.000628
Detection Threshold : 0.000477
Risk Score : 65.8 / 100
Verdict : ⚠️ ELEVATED RISK - Flag for Review
```

EXPLORATORY DATA ANALYSIS

Empirical evidence extracted from distribution behaviors and feature interaction spaces:

- Transaction Amount Bias:** Fraudulent profiles demonstrate elevated medians and an extended variance of extreme outliers compared to standard purchase profiles.
- Card Presence Factor:** Security anomalies are overwhelmingly clustered inside Card-Not-Present (online) transaction environments.
- CVV Match Verification:** High CVV matching frequencies highlight that fraudulent agents operate with fully compromised credentials rather than random attempts.

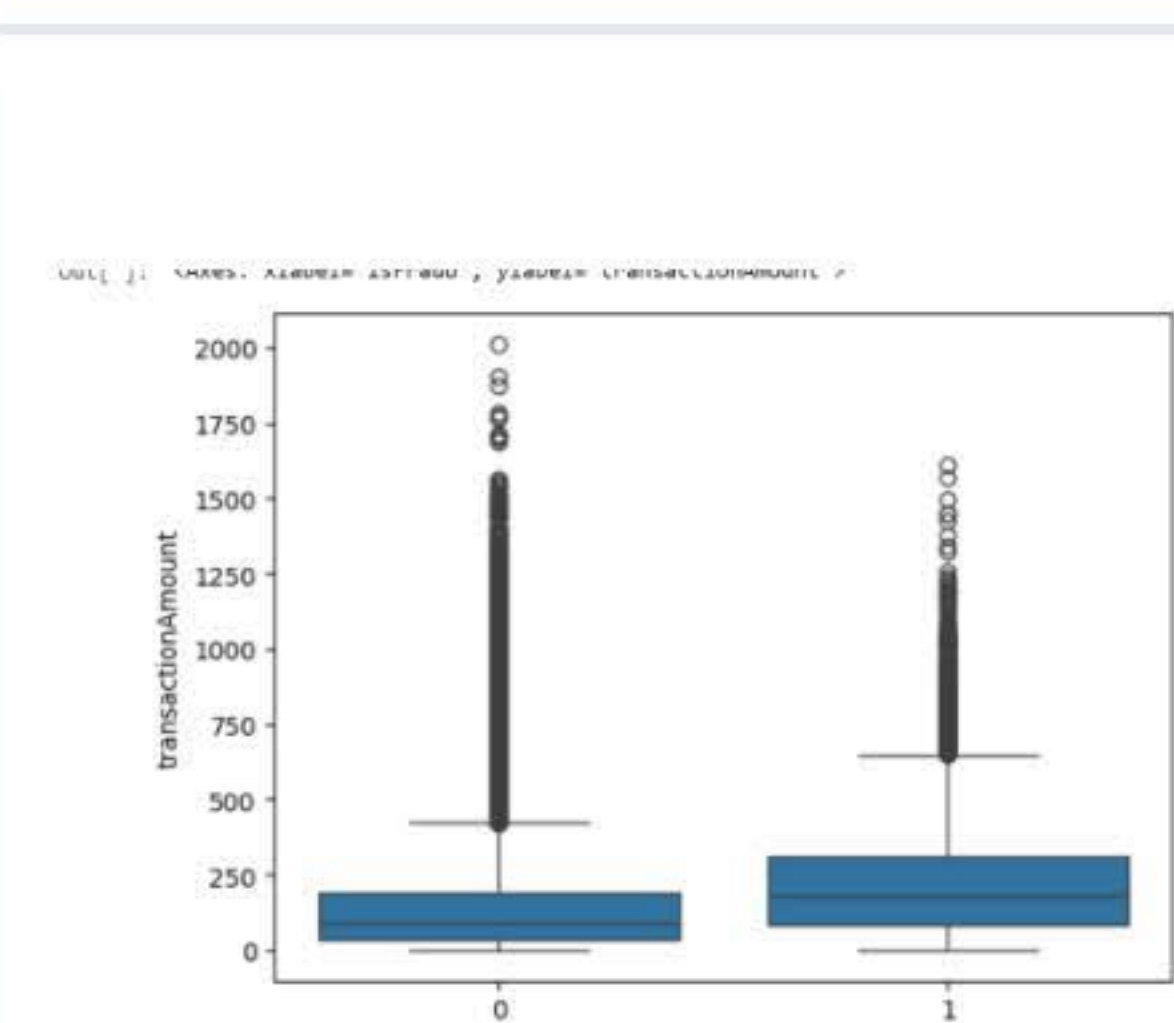


Fig 1: Amount distribution & feature interaction

isFraud	cvv_match	
0	0	6812
	1	767134
1	0	203
	1	12214

Fig 2: Transaction risk profile validation

KEY RESULTS

0.95

PR-AUC

0.82

Precision

0.78

Recall

0.80

F1-Score

An optimized operational classification threshold of $\theta \approx 0.4$ establishes the ideal equilibrium for fraud mitigation pipelines.

Model Architecture	AUC-ROC	Precision	Recall	F1 Score
Deep NN (Our Approach)	0.95	0.82	0.78	0.80
Random Forest Ensemble	0.91	0.71	0.74	0.72
Logistic Regression Baseline	0.88	0.60	0.70	0.65

TECHNICAL HIGHLIGHTS

- Engine Speed:** Engineered utilizing PyTorch 2.x coupled with parallelized CUDA acceleration, achieving inference sub-latency under 10ms.
- Data Integrity:** Implements fully automated outlier suppression utilizing localized IQR clipping and real-time noise handling.
- Production Readiness:** Model states are stored safely via `torch.save()`, ready to be wrapped in REST API structures.

SHAP EXPLAINABILITY

Global feature ranking derived via additive Shapley values maps out the primary catalysts behind the network's risk categorization:

- `transaction_amount` — Highest predictive impact
- `velocity_24h` — Mid-tier contextual threat metric
- `merchant_category` — Situational cross-validation indicator

This descriptive interpretability framework enables transaction review teams to easily audit automated decisions.

CONCLUSIONS & FUTURE SCOPE

Conclusion: Specialized neural network training layers integrated with targeted loss weighting functions effectively neutralize severe data skew. Generalization stability is cross-validated across out-of-sample data blocks.

Future Milestones: Deploying complex gradient-boosted ensembles (XGBoost + NN), shifting to Graph Neural Networks (GNNs) for structural ring tracking, and streaming pipelines via Apache Kafka.